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Roll No. _____

REF NO. 058472

B. Sc. (Hons) Semester IV Examination

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MATHEMATICS

Paper No. MTB-401 : Partial Differential Equations

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.

Answer five questions including Question 1 which is compulsory.

1. (a) Eliminate arbitrary constants α and β from $z = (x - \alpha)^2 + (y - \beta)^2$ to form a partial differential equation. **2**
- (b) Obtain the partial differential equation governing the equations $\phi(u, v) = 0$, $u = xyz$, $v = x + y + z$. **2**
- (c) For the first-order partial differential equation, explain the following terms and give example for each term:
General solution; Complete solution; Singular solution. **3**
- (d) Transform the partial differential equation $x^2p^2 + y^2q^2 = z^2$ into another partial differential equation by using the transformations $X = \log x$, $Y = \log y$, $Z = \log z$. What is the nature of new partial differential equation? **3**
- (e) Let $f(p, q) = 0$ be the first-order partial differential equation, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. Show that—
 - (i) Charpit's equations reduce to $p = \alpha$, where α is a constant;
 - (ii) for $q = Q(\alpha)$, the solution of the partial differential equation is $z = \alpha x + Q(\alpha)y + \beta$, where β is a constant. **3**
- (f) Define Clairaut type first-order partial differential equation and find its complete integral. **3**

(3)

2. (a) Define isoperimetric problem. **2**
(b) State Beltrami identity in Calculus of variations. **2**
(a) State and prove 'sufficient conditions for the existence of Laplace transform'. Further, show that the condition is not necessary by giving an example. **6**
(b) Find the Fourier series of the periodic function of period 2π , given by

$$f(x) = \begin{cases} 0, & -\pi \leq x \leq 0 \\ x, & 0 \leq x \leq \pi \end{cases}$$

6

3. (a) Find the Laplace transform of $f(t) = \sinh at + \frac{1}{\sqrt{\pi t}} - \frac{e^{at}-e^{-at}}{n-1}t!$. **6**
(b) Show that—

$$\mathcal{L} \left\{ \frac{d^n f(t)}{dt^n} \right\} = s^n \mathcal{L}\{f(t)\} - f(0)$$

3

- (c) Find the extremal of the functional $I[y(x)] = \int_0^1 (y^2 + y'^2) dx$ with $y(0) = 0$ and $y(1) = 1$. **3**

(4)

4. (a) State convolution theorem for Laplace transform. Using convolution theorem, evaluate

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2(s^2 - a^2)} \right\}$$

6

- (b) Using Laplace transform, show that—

$$\int_0^\infty \frac{\sin t}{t} dt = \frac{\pi}{2}$$

6

- (c) Define half-range sine series and cosine series.

2

5. (a) Using Laplace transform, find the solution of $y'' + 2y' + y = 0$ with $y'(0) = -2$ and $y(1) = 2$.

6

- (b) If

$$f(x) = \begin{cases} mx, & 0 < x < 1 \\ \frac{m}{2}(2 - x), & 1 < x < 2 \end{cases}$$

then find the half-range cosine series of $f(x)$ and show that—

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots = \frac{\pi^4}{96}$$

6

(5)

6. (a) Derive Euler-Lagrange equation for finding the extremal of the Functional. **6**
(b) Find the extremal of the functional $I[y(x)] = \int_0^{\pi/2} (y'^2 - y^2 - 2y \cos x) dx$ with $y(0) = 0$ and $y(\pi/2) = 0$. **6**

7. (a) Is Jacobi condition fulfilled for the extremal of the functional

$$\int_0^a (y'^2 + y^2 + x^2) dx$$

with $y(0) = 0, y(1) = 0$? Explain it. **6**

- (b) Find Laplace transform of

$$f(t) = \begin{cases} t^2, & 0 \leq t \leq 1 \\ \sin 2t, & 1 < t \leq \pi \\ \cos t, & t > \pi \end{cases}$$

using the unit-step function. **6**
